1. Identity

Document	STAR-PKG-STAR-SIDEREAL-EN · v2.2 · 2026-11-02
Package	star-sidereal
Version	0.1.2
Capability	star.time.sidereal
Maturity	Released & Verified (RELEASED)
Python	⩾3.9
Source commit	dc09f61d
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_sidereal

2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_sidereal-0.1.2-py3-none-any.whl
```

FILE	SHA-256
star_sidereal-0.1.2-py3-none-any.whl	15bcf7968a126b6608881b997a179297d783489493852a4e32bf5f3ee07bddbf
star_sidereal.py	adb187e78d0b9f2a7a7150f3cb385611596c71d7ce9325e7ff526dd8c94a9d3e
star_sidereal-0.1.2-source.zip	e2763f25c91033cbbe93ce6b970bd1cc75173c684d2970b8cb64778de76c03f1

5. What it does, and its limits

GMST and local mean sidereal time, IAU 1982 (Vallado eq. 3-47); verified against Vallado and ERFA.

IAU 2000/2006 (Earth rotation angle based) sidereal time, apparent sidereal time (equation of the equinoxes), UTC→UT1.

- `gmst82_deg(jd_ut1_day, jd_ut1_frac=0.0)` → degrees in [0, 360), Vallado eq. 3-47 (Aoki et al. 1982).
- `lst_deg(jd_day, jd_frac, east_longitude_deg)` → degrees in [0, 360).
- Non-finite input raises `ValueError`. Input is UT1 (not UTC): the caller applies DUT1.

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_star_sidereal_0_1_2,
  author  = {Cavazzini, Andrea},
  title   = {{star-sidereal}: Greenwich mean sidereal time (IAU 1982)},
  version = {0.1.2},
```

```
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_sidereal/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit dc09f61d}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

Related: space activities and export control · acceptable use · evidence and limits · licence texts